Design, Analysis and Test Plan of Lightweight Tubular Engine-Mount Strut for a Reusable Launch Vehicle

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Abstract

This project develops a lightweight engine-mount strut for a reusable launch vehicle by coupling parametric optimization with a strict structural hierarchy. Derived from a Merlin-class propulsion architecture via a geometric scale factor, a two-variable Python sweep over tube dimensions identifies an optimal $44\ \mathrm{mm} \times 2.5\ \mathrm{mm}$ 6061-T6 aluminum geometry, achieving a 36% mass reduction over a solid bar. Since the strut's low slenderness ratio of 27.2 falls below the critical transition threshold, the Johnson parabolic formula is used rather than Euler equations to reflect realistic inelastic yielding. Additionally, end fittings are sized to $1.25\times$ the strut's capacity to guarantee that column buckling and mid-length yielding govern failure prior to any joint or pin degradation. Failure is further determined by the load eccentricity. The report specifies the FEA, the instrumented test to failure, the correlation metrics and the post-test redesign method that will establish how accurately hand calculation and FEA predict failure.

Key Results and Engineering Contributions

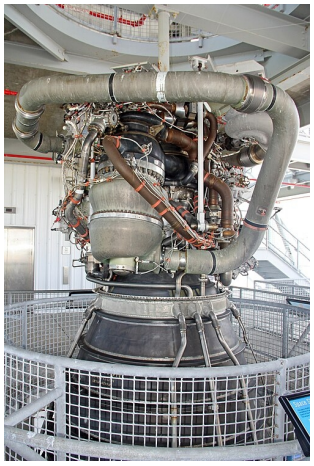
- **Mass optimization:** Reduced calculated shank mass by 36% relative to the lightest solid round bar meeting the same analytical criteria.
- **Structural sizing:** Selected a 44 mm OD × 2.5 mm wall 6061 T6 aluminum tube using a Python based parametric sweep of diameter and wall thickness.
- **Strength and stability:** Calculated yield and ultimate margins of +0.30 and +0.40 at the factored design loads, with a +0.12 buckling margin against a 10% requirement.
- **Eccentric loading:** Predicted a 70 to 83 kN failure load across 0.5 to 2 mm load eccentricity, identifying sensitivity to alignment as a key design consideration.
- **Controlled failure sequence:** Sized the end fittings to keep joint failure from preceding the predicted column failure, supporting meaningful model to test correlation.
- **Test and redesign framework:** Defined the FEA outputs, instrumentation, correlation metrics, and post test redesign criteria before physical testing.

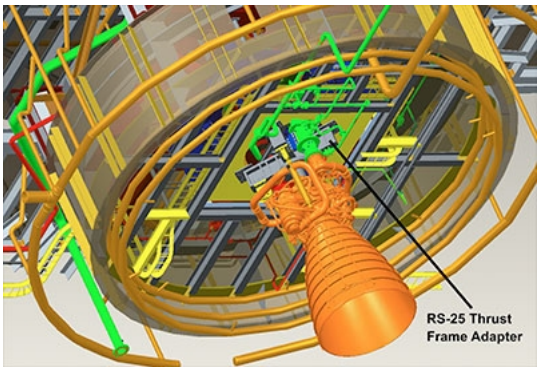
1. Introduction

1.1 Background

A launch-vehicle engine mount must carry thrust, resist engine gimbal and inertial loads, and weigh as little as possible. Across the industry, vehicles connect engines to structure in a few recurring ways:

1. The gimbal-bearing pedestal (SLS/RS-25, Shuttle). Each RS-25 attaches to the vehicle through a titanium gimbal bearing — a universal ball-and-socket joint, bolted to the launch vehicle by its upper flange and to the engine by its lower flange, supporting 3,390 kg of engine weight and over 500,000 lbf of thrust while pivoting ±10.5°.





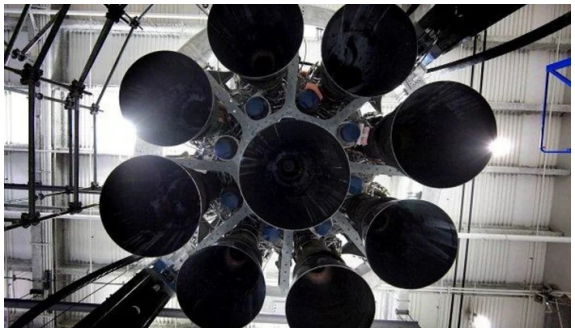
2. The strut/fitting truss ("thrust cone") — Saturn V F-1, and the internal structure inside modern thrust sections. Engines hang from a conical or ring frame through pin-jointed struts and machined end fittings. This is why the F-1 cluster photos show a conical lattice above the nozzles, and SLS CAD shows an "RS-25 Thrust Frame Adapter."

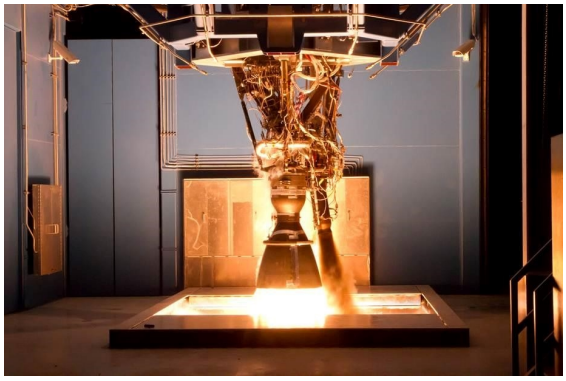




3. The ring-frame cluster (SpaceX Octaweb). Eight Merlins in a circle around a center engine on a machined aluminum structure that "reduces the length and weight of the Falcon 9 thrust structure and

streamlines manufacturing." Look at the base of the booster: the engines bolt into machined pockets of one monolithic aluminum frame.





4. Hold-down/release fittings. At the pad, the vehicle is restrained against release mechanisms (visible as the arm-like fittings gripping the Falcon 9 base below) — a deliberately fusible, high-load joint that is its own whole discipline.



The common thread: every one of these is built from **two-force members (struts) + pinned lug/clevis fittings + a frame**. A strut is a good test article because every failure mode has a closed-form prediction, and it can be broken in an ordinary compression test.

1.2 Research question and objectives

Research question. How can a lightweight engine support attachment meet a defined strength and stiffness requirement while minimizing mass, and how accurately can hand calculations and FEA predict its actual failure?

Objectives:

1. Define a load case and requirements traceable to published sources.
2. Size a minimum-mass strut that meets the strength and stability criteria, using a two-variable sweep.
3. Design end fittings so that the strut column, not the joint, is the first element to fail.
4. Predict the stiffness, the failure load and the failure mode by hand calculation and FEA.
5. Test the part to failure, quantify the prediction error, and redesign from the observed failure mechanism.

1.3 Summary of results

The selected strut is a tube of 44 mm outside diameter and 2.5 mm wall, 300 mm long, with 56 mm diameter solid ends tapering into 15 mm thick tongues drilled for 20 mm pins. Pin-to-pin length is 400 mm and the mass is 0.805 kg. At the design limit load of 42.5 kN the secant peak stress is 146 MPa. Margins at 1.25 and 1.4 times limit load are +0.30 and +0.40, and the buckling margin is +0.12 against a 10% requirement. The predicted peak load is 70 to 83 kN, and the ends are sized to carry at least 1.25 times the strut's predicted maximum.

2. Strut concept and configuration

2.1 Definitions

The **strut** (the test specimen) is a single machined part with a hollow tubular shank, two solid tapered transitions and two flat **tongues** at its ends. Each tongue has a reamed hole for a pin. The **fixture** is a steel clevis at each end: two fork plates and a **pin** in double shear. The tongue is the inner lug and the fork plates are the outer lugs of the double-shear joint. The pin axis passes through the tongue thickness. Rotation about the pin axis is free, so the shank is pinned in the plane perpendicular to the pin axis. The forks restrain rotation about the other axes, and the analysis conservatively treats both planes as pinned.

This arrangement keeps the pin joint out of the column section. The tongue is wider than the tube wall can carry as a column, the hole sits in a short solid region that cannot buckle as a column, and the shank is sized purely by column behaviour. There is no second joint between the tongue and the shank, because they are one part.

2.2 Load path and free-body diagrams

Figure 1 shows the load path. Engine thrust T is shared by N inclined struts. Each strut is a two-force member: the force is equal and opposite at its two pins and acts along the line joining them. In the ideal case the strut carries a pure axial load. In practice the load line is offset from the centroid by an eccentricity e, and the strut bends as it is loaded, which is analysed in a section on buckling. At the pin joint, the strut tongue transmits P to the pin and the pin transmits P/2 to each fork plate. The pin bending arm in this joint governs the pin design.

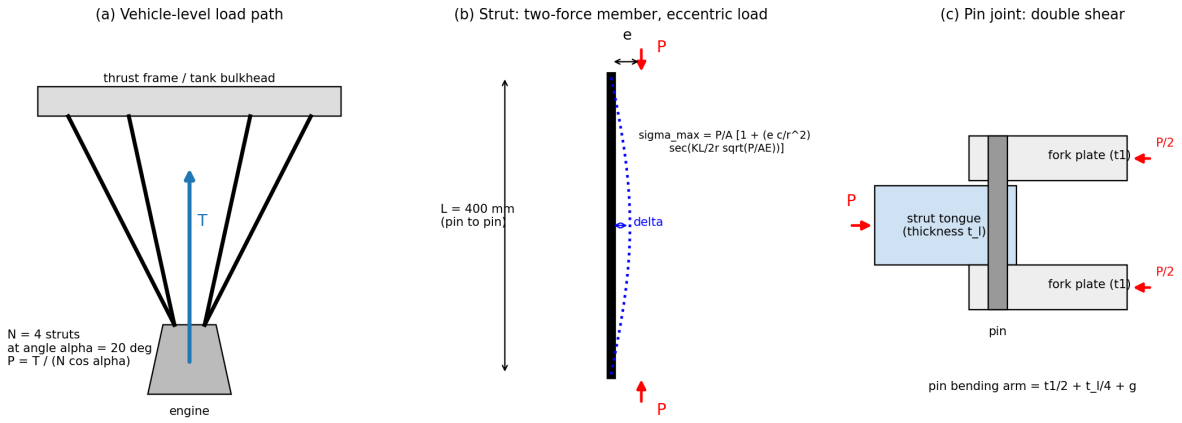


Figure 2 shows the strut geometry.

2.3 Failure-order design rules

Since the test is intended to answer the research question, the failure sequence is controlled. The strut column should fail first, and the end hardware should remain elastic. Four design rules enforce this:

1. The shank meets all strength and stability criteria of Section 3.3 at design-minimum properties.
2. The tongues, holes and pins carry at least 1.25 times the strut's predicted maximum capacity, the Johnson column load at typical properties.
3. The pins stay elastic up to 1.10 times that maximum, so pin bending does not alter the measured load eccentricity.
4. The fixture and test machine are rated for at least 1.5 times that maximum.

3. Requirements and design basis

3.1 Material and pin data

Quantity	Value	Source
6061-T6 elastic modulus E	68.9 GPa	ARRK datasheet [8]
Shear modulus G	26 GPa	[8]
Density	2.70 g/cm3	[8]
Yield / ultimate strength, typical	276 / 310 MPa	[8], MatWeb extract [9]
Yield / ultimate strength, design minimum	240 / 290 MPa	6061-T6 minimums [10]
Shear strength	207 MPa	[8]
Bearing ultimate / yield (edge distance / diameter = 2)	607 / 386 MPa	[9]
Pin steel (4130): Ftu, Fty, Fsu	861.8, 710.2, 565.4 MPa	Values printed in the worked example of the Air Force Stress Manual [14]
Thrust and diameter, Merlin 1D (sea level)	845 kN, 0.92 m	Wikipedia [1]
Ultimate factor of safety	1.4	NASA-STD-5001B [7]
Yield factor of safety	1.25	[7] (protoflight value)

Design-minimum strengths are used for all pass/fail criteria. Typical strengths are used only to predict the most probable test outcome.

3.2 Requirements

ID	Requirement
R1	Maximum secant stress at 1.25 x limit load does not exceed the design yield strength (240 MPa)
R2	Maximum secant stress at 1.4 x limit load does not exceed the design ultimate strength (290 MPa)
R3	The column critical load at design-minimum properties is at least 1.10 x the ultimate load (buckling margin of at least 10%, consistent with published launch-structure practice [23])
R4	Tube wall: D/t at most 20, t at least 2.0 mm, and classical local buckling stress at least 3 x design yield
R5	Tongues, holes and pins carry at least 1.25 x the strut's predicted maximum; pins elastic to 1.10 x
R6	Axial stiffness is measured and reported
R7	Predicted peak load is within 15% of the measured value when the prediction uses measured eccentricity and material strength

3.3 Modelling assumptions

Assumption	Value	Basis
Number of struts N	4	Representative of a four-strut mount
Angle α from the thrust axis	20 degrees	Modeling assumption
Full-scale strut length	Equal to engine diameter (920 mm)	Sets the geometric scale factor
Test strut length L (pin centre to pin centre)	400 mm	Bench-test scale
End factor K	1.0	Pinned ends, conservative for the plane the forks restrain
Load eccentricity e	1.0 mm	Tolerance stack (Section 10.3); sensitivity 0.5 to 5 mm
Maximum tube outside diameter	50 mm	Envelope assumption for the sweep
Wall slenderness and minimum wall	D/t at most 20, t at least 2.0 mm	Robustness to machining and handling
Local buckling criterion	Classical stress at	Allows a shell knockdown factor down to one

Assumption	Value	Basis
	least 3 x design yield	third; see Section 5.4
Fork thickness and pin gap	10 mm each and 0.5 mm per side	Fixture design
Lug coefficients K_n and K	0.74 and 1.33	Lowest values in the Air Force manual's worked examples; to be replaced from its figures

4. Load definition

Define a dimensionless geometric scale factor, λ. The outer diameter of the engine (0.92m for the Merlin 1D) is publicly known, standardized dimension, it is taken to be the characteristic dimension. λ is defined as the ratio of test strut length L to engine diameter D_{eng}:

$$\lambda = \frac{L}{D_{eng}} = \frac{400}{920} = 0.4348, \qquad \lambda^2 = 0.1890$$

Four struts inclined at 20 degrees to the thrust axis share the thrust T. Resolving force balance along the thrust axis:

$$P_{full} = \frac{T}{N \cos \alpha} = \frac{845{,}000}{4 \times 0.9397} = 224{,}800 \text{ N}$$

Geometric similarity keeps stress constant if force scales with area, i.e. with λ squared:

$$P_L = \lambda^2 P_{full} = 0.1890 \times 224{,}800 = 42{,}500 \text{ N}$$

The factored loads are the yield load P_y = 1.25 P_L = 53.1 kN and the ultimate load P_u = 1.4 P_L = 59.5 kN. The load case is axial thrust only. Gimbal side load and engine inertia are outside the scope of this test and are treated as extensions (Section 12).

5. Analytical methods

5.1 Tube section properties

For a round tube of outside diameter D and wall t, with inside diameter d = D - 2t:

$$A = \frac{\pi}{4}\left(D^2 - d^2\right), \qquad I = \frac{\pi}{64}\left(D^4 - d^4\right), \qquad r = \sqrt{I/A}, \qquad c = \frac{D}{2}$$

The round section has the same radius of gyration in every plane, so there is a single buckling plane to check.

5.2 Global buckling

With slenderness S = K L / r, the Euler load and the transition slenderness are:

$$P_E = \frac{\pi^2 E I}{(K L)^2}, \qquad C_c = \sqrt{\frac{2\pi^2 E}{F_{ty}}}$$

Below C_c the Johnson parabola applies, and above it Euler applies:

$$\sigma_{cr} = F_{ty} - \frac{F_{ty}^2}{4\pi^2 E} S^2 \quad (S < C_c), \qquad \sigma_{cr} = \frac{\pi^2 E}{S^2} \quad (S \geq C_c), \qquad P_{cr} = \sigma_{cr} A$$

C_c is 75.3 at the design yield strength and 70.2 at the typical yield strength.

5.3 Eccentric loading

The load line is offset from the centroid by e. The secant formula gives the maximum stress and the additional lateral deflection at mid-length ([11]):

$$\varphi = \frac{K L}{2r}\sqrt{\frac{P}{A E}}, \qquad \sigma_{max} = \frac{P}{A}\left[1 + \frac{e\,c}{r^2}\sec\varphi\right], \qquad \delta_{max} = e\,(\sec\varphi - 1)$$

The formula is valid for pinned-pinned columns [12]. For a column that yields before it buckles, the useful failure indicator is the load at which σ_{max} reaches the yield strength. That load is always below the critical load.

5.4 Local buckling

The classical axial buckling stress of a thin cylindrical shell of mean radius R and wall t is 0.605 E t / R, reduced in practice by an imperfection knockdown factor γ (NASA SP-8007 method [16], formula summary [24]):

$$\sigma_{cl} = 0.605\,\gamma\,\frac{E\,t}{R}, \qquad \gamma = 1 \text{ for the classical value}$$

The criterion adopted here requires the classical value to exceed three times the design yield strength, which tolerates a knockdown factor down to one third. For the selected tube it exceeds the yield strength by a factor of 20.9, so local buckling does not govern.

5.5 Pin joint

The tongue-and-pin joint is analysed with the double-shear equations of the Air Force Stress Manual [14]. If e_h is too small, the high bearing load from the pin can shear or tear the material out right through the end of the tongue. This shear-out or net-section failure only happens in tension.

Here D is the hole and pin diameter, t_l the tongue thickness, e_h the distance from hole centre to tongue end, w the tongue width, t_1 the thickness of each fork plate, and g the gap per side

Bearing Capacity P_{brg} calculates how much crushing load the cylindrical wall of the hole can withstand before the pin permanently deforms or crushes the tongue material over $\mathrm{D} \times t_l$, the projected bearing area).

$$\mathrm{P}_{brg} = \min(K F_{tu},\ 1.304 K F_{ty},\ F_{bru},\ 1.304 F_{bry})\,D\,t_l$$

Net-Section Capacity P_{net}: when the pin pulls on the tongue, it creates tension across the remaining "meat" of the material on either side of the hole. The effective width is the total tongue width minus the hole diameter $(\mathrm{w} - \mathrm{D})$, multiplied by the tongue thickness t_l.

$$P_{net} = K_n F_{tu} (w - D)\, t_l$$

Pin Structural Capacities $\mathrm{P}_{ub,pin}, P_{el,pin}, P_{us,pin}$:

$\mathrm{P}_{ub,pin}$ (Ultimate Bending): The maximum bending moment the steel pin can handle before failing.

$\mathrm{P}_{el,pin}$ (Elastic Limit): The load limit where the pin stays completely elastic without taking a permanent bend, which is set to $1.10\ \times$ the strut's max capacity.

$\mathrm{P}_{us,pin}$ (Double Shear): The load required to snap the 20 mm steel pin cleanly across its two shear planes ($1.571 D^2 F_{su,P}$).

$$b_{arm} = \frac{t_1}{2} + \frac{t_l}{4} + g$$

$$P_{ub,pin} = \frac{0.1963\, k_b\, D^3 F_{tu,P}}{b_{arm}}$$

$$P_{el,pin} = \frac{0.1963\, D^3 F_{ty,P}}{b_{arm}}$$

$$P_{us,pin} = 1.571\, D^2 F_{su,P}$$

The bearing and net-section equations apply when e_h / D is at least 1.5, and the plastic bending coefficient k_b is 1.56 for a ductile pin. The manual folds shear tear-out into the bearing coefficient K.

5.6 Compression loading of a tension-formulated lug

The manual's lug equations describe lugs loaded in tension. In compression the pin bears on the shank side of the hole, and the material between the hole and the free end is unloaded, so tear-out toward the free end is not the expected mode. No source found for this work gives compression allowables. The tension equations are therefore used as a conservative bound, which also covers load reversal in service. The test applies compression only.

6. Shank sizing

6.1 Sizing Methodology and Design Space Definition

To identify the minimum-mass tubular shank, a two-variable parametric sweep was conducted over the outer diameter (D) and wall thickness (t).

Design Space Boundaries:

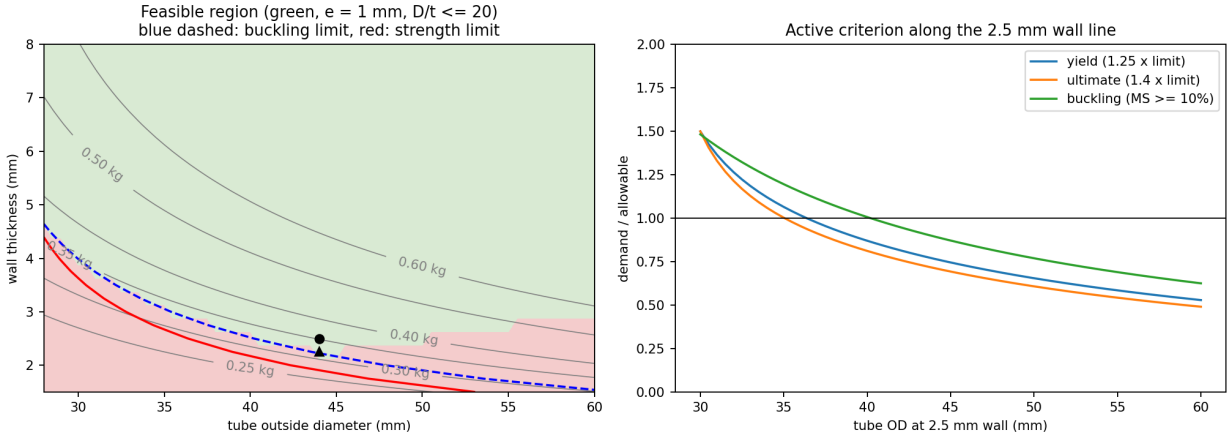
Outside Diameter $(28 \text{ to } 60 \text{ mm})$: The lower bound (28 mm) starts just above the lightest solid bar diameter (25.5 mm) to ensure it evaluates viable hollow alternatives. The upper bound (60 mm) is

chosen to be comfortably larger than any reasonable optimum, ensuring the search space is fully bracketed.

Wall Thickness $(1.5 \text{ to } 12 \text{ mm})$: 1.5 mm is chosen as a lower bound approaching the minimum practical wall thickness for machining and handling without risking burn-through or excessive local shell buckling. 12 mm is an intentionally thick upper bound that approaches a solid bar.

Packaging and Manufacturing Constraints: To respect vehicle-level spatial clearance and prevent geometric interference within the engine bay, a strict upper envelope limit of $D \leq 50 \text{ mm}$ was enforced. Additionally, a minimum wall thickness constraint $(t \geq 2.0 \text{ mm})$ and a slenderness ceiling $(D/t \leq 20)$ were implemented to guarantee structural robustness against handling damage, tolerance stack-ups during deep-hole boring, and premature local shell buckling.

Evaluation Logic: For every valid coordinate pair (D, t), the model evaluates structural criteria R1 through R4 at the design-minimum material properties. Feasible configurations are filtered, and the global optimum is identified by selecting the geometry that minimizes cross-sectional area over the fixed 400 mm pin-to-pin length. Figure 3 shows the feasible region and the active criteria.



6.2 Results and sensitivity

With D/t at most 25 and t at least 2.0 mm:

e (mm)	Maximum OD (mm)	Optimum OD x wall (mm)	Area (mm2)	Mass over 400 mm (kg)	Governing criterion
0.5, 1.0 or 2.0	40	38.0 x 2.75	304.5	0.329	Buckling margin
0.5, 1.0 or 2.0	50 or 60	48.0 x 2.00	289.0	0.312	Buckling margin
5.0	40	39.5 x 3.50	395.8	0.428	Yield
5.0	50 or 60	50.0 x 2.25	337.5	0.365	Yield

With e = 1 mm and OD at most 50 mm, the optimum depends on the wall-slenderness bound:

D/t limit	Optimum OD x wall (mm)	Area (mm2)	Mass over 400 mm (kg)	Buckling utilization
15	38.0 x 2.75	305	0.329	0.98
20	44.0 x 2.25	295	0.319	0.99
25 or 30	48.0 x 2.00	289	0.312	1.00

Conclusions:

- Column Buckling Governs Design Limits: As illustrated in Figure 3, global column buckling (Constraint R3) forms the active constraint boundary for all realistic load eccentricities ($\mathrm{e} \leq 2.0\ \mathrm{mm}$). Material yield only governs under extreme misalignments ($\mathrm{e} \geq 5.0\ \mathrm{mm}$). Because buckling dominates within the expected tolerance range ($0.5\ \mathrm{to}\ 2.0\ \mathrm{mm}$), the theoretical optimum mass remains essentially invariant at approximately $0.32\ \mathrm{kg}$.
- Low Mass Sensitivity to Wall Slenderness $(\mathrm{D/t})$: evaluating the feasible region shows that relaxing the wall-slenderness constraint $(\mathrm{D/t})$ from 15 to 25 yields only a $5\%(17\ \mathrm{g})$ mass reduction (decreasing shank mass from $0.329\ \mathrm{kg}$ to $0.312\ \mathrm{kg}$). Because structural efficiency is insensitive to wall thickness within this regime, selecting a conservative ratio ($\mathrm{D/t} = 17.6, \mathrm{t} = 2.5\ \mathrm{mm}$) imposes a negligible mass penalty while significantly improving manufacturing robustness and deep-hole boring tolerance.
- Immunity to Local Shell Buckling: across all feasible sweep coordinates, the classical local shell-buckling utilization remains below 0.20, providing a margin of safety greater than 4.0. Localized wall crimping is therefore non-governing, confirming that wall thickness selection can be dictated entirely by handling durability and global column stability.

6.3 Selected shank

The optimum for D/t at most 20 is 44.0 x 2.25 mm. The wall is rounded up to 2.5 mm so that a bore tolerance of the order of 0.1 mm does not remove a significant fraction of the wall. The selected section is a tube of 44.0 mm outside diameter, 2.5 mm wall and 39.0 mm inside diameter:

$$A = 325.9\ \mathrm{mm}^2, \quad I = 70{,}423\ \mathrm{mm}^4, \quad r = 14.699\ \mathrm{mm}, \quad S = \frac{400}{14.699} = 27.21, \quad C_c = 75.28$$

Since S is below C_c, the Johnson branch applies. The Euler load (299 kN) corresponds to a stress of 918 MPa, which exceeds the yield strength and is not physical for this stocky column.

$$\sigma_{cr} = 240 - \frac{240^2}{4\pi^2(68{,}900)}(27.21)^2 = 224.3\ \mathrm{MPa}$$

$$P_{cr} = 224.3 \times 325.9 = 73.1\ \mathrm{kN}, \qquad MS = \frac{73.1}{1.1 \times 59.5} - 1 = +0.12$$

At the limit load (42.5 kN):

$$\varphi = \frac{400}{2(14.699)}\sqrt{\frac{42{,}500}{325.9 \times 68{,}900}} = 0.5919$$

$$\sec\varphi = 1.2050$$

$$\delta_{max} = 0.205 \text{ mm}$$

$$\sigma_{max} = 130.4\left[1 + \frac{(1.0)(22)}{14.699^2}(1.2050)\right] = 146.4 \text{ MPa}$$

Load case	P (kN)	P/A (MPa)	sec	Peak stress (MPa)	Allowable (MPa)	Margin
Limit	42.50	130.4	1.2050	146.4	n/a	n/a
1.25 x limit	53.12	163.0	1.2676	184.0	240	+0.30
1.4 x limit	59.50	182.5	1.3078	206.8	290	+0.40

The wall slenderness is D/t = 17.6, and the classical local-buckling stress is 5,022 MPa, 20.9 times the design yield strength.

6.4 Comparison with a solid bar

The lightest solid round bar that meets R1 to R3 has a diameter of 25.5 mm (511 mm^2 area, 0.552 kg over 400 mm). The selected tube has 0.352 kg over the same length, 36% lighter, and the sweep optimum (0.319 kg) is 42% lighter.

7. End-fitting design

7.1 Requirements

The strut's predicted maximum capacity is the Johnson load at typical properties, 83.2 kN. Rules 2 and 3 of Section 2.3 give:

$$P_{end} = 1.25 \times 83.2 = 104.0 \text{ kN}, \qquad P_{pin,elastic} = 1.10 \times 83.2 = 91.5 \text{ kN}$$

7.2 End-Fitting Selection

End fittings are sized to carry at least $1.25\times$ the strut's predicted maximum capacity (104.0 kN).

Python is again used to evaluate the end fittings using the following specific increments/ranges:

Head bar stock diameters (D_h): Discrete set of standard sizes: $50{,}56{,}60$, and 63.5 mm.

Pin diameters (D_p): Discrete set of sizes: $16{,}18{,}20{,}22{,}25{,}28{,}$ and 30 mm.

Tongue thicknesses (t_l): Swept from 10 mm up to $0.8 \times D_h$ using an increment of 1.0 mm

The selected configuration, optimized for minimum volume, utilizes a 56 mm stock diameter, a 20 mm pin, and a 15 mm thick tongue.

Head Stock Selection and Net-Section Justification

The tongue is formed by milling two flats on the bar, so its width is $w = \sqrt{D_{head}^2 - t_l^2} = \sqrt{56^2 - 15^2} = 54.0\ \text{mm}$.

The edge distance is 1.6 pin diameters rounded up to the next 5 mm. $1.6 \times D_{pin} = 1.6 \times 20\ \text{mm} = 32\ \text{mm}$. Rounding up to the next $5\ \text{mm}$ gives $e_h = 35\ \text{mm}\ (e_h/D = 1.75)$

End fittings require a minimum capacity of $P_{end} = 104.0\ \text{kN}$ to ensure the column fails first. Machining tongues directly from $44.0\ \text{mm}$ tube stock ($t_l = 15.0\ \text{mm},\ D_p = 20.0\ \text{mm}$) yields a chord width of $w = \sqrt{44.0^2 - 15.0^2} = 41.36\ \text{mm}$. This produces a net-section capacity of $P_{net} = K_n F_{tu}(w - D_p)t_l = 0.74 \times 290 \times (41.36 - 20) \times 15 = 68.9\ \text{kN}$, which fails the $104.0\ \text{kN}$ requirement. Heads are therefore turned from a larger $56.0\ \text{mm}$ bar stock ($w = 54.0\ \text{mm}$), increasing capacity to $109.3\ \text{kN}$.

The result:

Parameter	Value
Head bar stock diameter	56 mm
Pin and hole diameter	20 mm (H7 hole, g6 pin)
Tongue thickness	15 mm
Tongue width (chord)	54.0 mm
Edge distance e_h (hole centre to end)	35 mm (e_h / D = 1.75, a = 25.0 mm)
Tongue zone length	55 mm
Solid taper from 56 mm to 44 mm diameter	30 mm
Tube length	300 mm
Overall length	470 mm (pin centre distance 400 mm)
Fork plates and pin gap	10 mm each, 0.5 mm per side

7.3 Capacities and margins

The pin bending arm is 10/2 + 15/4 + 0.5 = 9.25 mm. Capacities with design-minimum properties (Section 3.1):

Mode	Capacity (kN)	Required (kN)	Margin vs requirement	Margin at ultimate design load (59.5 kN)
Tongue bearing	115.7 (123.7 typical)	104.0	+0.11	+0.94
Tongue net section	109.3 (116.8	104.0	+0.05	+0.84

Mode	Capacity (kN)	Required (kN)	Margin vs requirement	Margin at ultimate design load (59.5 kN)
	typical)			
Pin double shear	355	104.0	+2.42	+4.97
Pin ultimate bending	228	104.0	+1.19	+2.84
Pin elastic limit	121	91.5	+0.32	+1.28 (against the 53.1 kN yield load)

A two-plane tear-out estimate gives a high consistency margin of $170\ \mathrm{kN}$, though shear-out is inherently covered by the manual's bearing coefficient. Using typical material properties, tongue bearing and net-section capacities exceed the strut's maximum predicted load by 1.49 times and 1.40 times, respectively, while the pin elastic limit provides a 1.45 times margin against the yield load.

7.4 Fit

The pin and hole utilize an H7/g6 sliding fit. For the $18\ \text{to}\ 30\ \mathrm{mm}$ size band, standard fit tables [19] specify a hole tolerance of $0\ \text{to}\ +21\ \mu\mathrm{m}$ and a shaft tolerance of $-20\ \text{to}\ -7\ \mu\mathrm{m}$, resulting in a controlled radial clearance of $7\ \text{to}\ 41\ \mu\mathrm{m}$. A second reference [20] confirms the H7 hole band and describes H7/g6 as the sliding fit. The pin is therefore a sliding fit, 19.980 to 19.993 mm in a hole of 20.000 to 20.021 mm. This mechanical fit ensures smooth assembly while contributing a negligible maximum of $20.5\ \mu\mathrm{m}$ to total load eccentricity.

7.5 Mass and stiffness

The total part mass is $0.805\ \mathrm{kg}$, broken down into $0.264\ \mathrm{kg}$ for the tube, $0.320\ \mathrm{kg}$ for the solid tapers, and $0.221\ \mathrm{kg}$ for the end tongues. Consequently, the heavy end regions account for 67% of the total mass. A segmented-bar model predicts an axial stiffness of $61.9\ \mathrm{kN/mm}$ ($56.1\ \mathrm{kN/mm}$ for the bare tube over the $400\ \mathrm{mm}$ span). Because this model excludes pin, hole, and test machine compliance, actual measured laboratory stiffness is expected to be slightly lower.

8. Predicted Behavior

First yield is the load at which the secant peak stress reaches the typical yield strength of 276 MPa. When the strut reaches its Johnson load before first yield, the Johnson load governs.

Eccentricity e (mm)	Peak load (kN)	Governing mechanism	Peak / limit load	Lateral deflection at limit load (mm)
0.5	83.2	Johnson load reached before first yield	1.96	0.102
1.0	78.4	First yield	1.85	0.205
2.0	70.2	First yield	1.65	0.410

First yield is defined as the load where secant peak stress reaches the typical yield strength of 276 Mpa. If the strut reaches its column buckling limit beforehand, that critical load governs.

The predicted peak failure load is $70 \text{ to } 83 \text{ kN}$ ($16 \text{ to } 19 \text{ kip}$). As shown in Figure 4, failure is expected to initiate via material yielding on the concave face at mid-length, followed immediately by rapid lateral bowing while all end components remain fully elastic. All proof test loads (42.5, 53.1, and 59.5 kN) sit safely below the lowest predicted first-yield threshold (70.2 kN). At an eccentricity of 1 mm, the first-yield load efficiency is 97 kN/kg.

9. Finite element analysis plan

The FEA predicts stiffness, buckling, first yield, peak load and the stresses at the holes and tapers, and is checked against the hand calculations and then the test.

Item	Specification
Geometry	Full strut including tapers, tongues and holes; pins as steel solids; forks as rigid or elastic steel
Symmetry	Mid-length plane only; eccentricity breaks symmetry in the bending plane
Elements	Quadratic hexahedra where possible, tetrahedra in the tapers and tongues
Contact	Pin to hole, frictional (coefficient 0.1 assumed), with 7 to 41 micrometre clearance
Loading	Fork displacement with free rotation about the pin axis
Step 1	Linear static at 42.5, 53.1 and 59.5 kN
Step 2	Linear (eigenvalue) buckling, first four modes
Step 3	Geometrically nonlinear, with an initial bow equal to the measured eccentricity (or 1 mm) and elastic-perfectly-plastic material at typical yield
Convergence	Three mesh densities; peak stress change of at most 5% away from singularities
Outputs	Load against lateral deflection to peak; stresses at hole edges and taper; pin bending; axial stiffness
Acceptance	Linear stiffness and buckling load within 10% of the hand calculation

10. Test plan

10.1 Facility and machine capacity

The planning load is 1.5 times the strut's predicted maximum: 1.5 x 83.2 = 125 kN (28 kip, using 1 kip = 4.448 kN). The USC Structures and Materials Research Laboratory [15] lists universal testing machines from 1 to 1,000 kip and servo-hydraulic actuators from 11 to 340 kip, with instrumentation for load, displacement and strain. The plan assumes a machine rated at least 50 kip (222 kN), which keeps the expected load well inside the load-cell range. The specific machine, the load-cell accuracy between 20 and 120 kN, and the vertical clearance (470 mm part plus forks plus load train, about 650 mm) are confirmed with the laboratory before machining.

10.2 Fixture

- Two steel fork plates per end, 10 mm thick each, with 20 H7 holes and a gap of at most 0.5 mm per side.
- Pin centres define the 400 mm length. Load passes through the pins, not through flat platens, so the pinned end condition of the model is preserved.
- At the planning load the pin bears on the forks at 125,000 / (20 x 20) = 312 MPa, so the fork steel is specified with a bearing yield well above that value.
- Fixture axial stiffness of at least 620 kN/mm, ten times the predicted strut stiffness of 61.9 kN/mm, so that fixture compliance does not alter the end condition.
- Pins are 20 g6, 4130 steel, retained by circlips. The test is terminated and the pins are inspected if the load exceeds 100 kN without a peak, because the pins are designed to remain elastic to 91.5 kN and have an elastic limit of 121 kN.
- A guard surrounds the specimen, because column failure releases stored energy suddenly.

10.3 Load eccentricity: budget and measurement

The analysis uses an eccentricity of 1.0 mm. A tolerance-stack estimate gives a smaller value: hole-position error of 0.05 mm, fork and fixture misalignment of about 0.2 mm, and pin clearance of at most 0.02 mm (Section 7.4). The worst-case sum is 0.27 mm and the root-sum-square is 0.21 mm, so the analysis value is conservative by a factor of about four. The test replaces this estimate with a measurement.

Four strain gauges at mid-length, spaced 90 degrees around the tube, give bending in two planes. For a pair of opposite gauges with strains e1 and e2, the bending strain and the mid-length moment are:

$$\varepsilon_{bend} = \frac{\varepsilon_1 - \varepsilon_2}{2}, \qquad M = \frac{E\,I\,\varepsilon_{bend}}{c}, \qquad M = P\,e\,\sec\varphi$$

with I = 70,423 mm4 and c = 22 mm for the selected tube. The last relation follows from Section 5.3, because the moment at mid-length is P(e + delta) = P e sec(phi). Reading the strains at a low load gives e directly, and fitting the secant curve to the measured load-deflection data gives a second, independent estimate.

10.4 Instrumentation and procedure

Instrumentation: four mid-length strain gauges; two gauges at each taper-to-tube shoulder; a lateral displacement sensor at mid-length; an axial extensometer between pin centres, which excludes machine compliance from the stiffness measurement; the machine load cell; and video.

Before the test the following are measured: tube outside diameter and wall thickness at both ends and at mid-length, straightness, hole positions and diameters, and the yield and ultimate strength of tensile coupons cut from the same stock. These measured values replace the assumed ones in the hand calculation and the FEA before the result is known.

The load is applied in displacement control in stages: a 2 kN seating load; 42.5 kN (limit) and unload; 53.1 kN (1.25 x limit) and unload; 59.5 kN (1.4 x limit) held for 60 s; then continued at a slow, quasi-static rate agreed with the laboratory until the peak and the post-peak load drop. All three proof levels are below the lowest predicted first yield of 70.2 kN, so the staged loading should not damage the part.

10.5 Specimen tolerances

The tolerances are chosen for a first machined part and are to be confirmed with the machine shop.

Feature	Nominal	Tolerance or note
Material	6061-T6 bar, 56 mm	Mill certificate required
Overall length	470 mm	plus or minus 0.2 mm
Pin-hole centre distance	400.00 mm	plus or minus 0.02 mm
Pin holes	20 mm	H7 (20.000 to 20.021 mm), Ra 1.6 micrometres, perpendicular to the tongue faces within 0.02 mm
Hole centre offset from strut axis	0	plus or minus 0.05 mm
Tongue thickness (milled flats)	15.0 mm	plus or minus 0.1 mm
Tongue zone length	55 mm	plus or minus 0.2 mm
Edge distance e_h	35 mm	plus or minus 0.1 mm
Solid taper	30 mm long, 56 mm to 44 mm	smooth, no tool marks transverse to the axis
Tube outside diameter	44.0 mm	plus or minus 0.1 mm
Tube bore	39.0 mm, 300 mm long	bored from both ends, meeting at mid-length; wall 2.5 mm plus or minus 0.1 mm

Feature	Nominal	Tolerance or note
Concentricity of bore and outside diameter		0.1 mm
Edges	break 0.5 mm	removes edge starters

10.6 Verification matrix

ID	Requirement	Verification	Predicted margin
R1	Yield at 1.25 x limit	Analysis; test shows no permanent set after unloading from 53.1 kN	+0.30
R2	Ultimate at 1.4 x limit	Analysis; 60 s hold at 59.5 kN	+0.40
R3	Buckling margin of at least 10%	Analysis; eigenvalue FEA	+0.12
R4	Wall D/t at most 20, t at least 2.0 mm, local buckling stress at least 3 x yield	Analysis; measured wall	D/t = 17.6; local stress 20.9 x yield
R5	Ends at least 1.25 x strut maximum; pins elastic to 1.10 x	Analysis; post-test inspection	+0.05 (tongue net section, the tightest); +0.32 (pin elastic)
R6	Axial stiffness measured	Extensometer between pin centres	Predicted 61.9 kN/mm (bar model)
R7	Peak load within 15% of prediction made with measured e and strength	Correlation table	To be determined by the test

11. Correlation and redesign method

11.1 Correlation

The predictions in Section 8 are fixed before the test. After the test, each row of the table below is completed with the hand calculation, the FEA and the measurement, and the percent error is reported as (measured - predicted) / predicted.

Quantity	Hand calculation	FEA	Measured	Error
Axial stiffness (kN/mm)	61.9 (bar model)	to be completed	to be completed	to be completed
Load eccentricity e (mm)	1.0 assumed	not applicable	from strain gauges	to be completed

Quantity	Hand calculation	FEA	Measured	Error
Lateral deflection at 42.5 kN (mm)	0.205	to be completed	to be completed	to be completed
First yield (kN)	70.2 to 83.2	to be completed	from load-strain curve	to be completed
Peak load (kN)	70.2 to 83.2	to be completed	to be completed	to be completed
Failure mode and location	Yield at concave face, then bow	to be completed	to be completed	match or not

The correlation is repeated with the measured eccentricity and measured coupon strength, which separates input uncertainty (the eccentricity and the strength) from model error (the formulas).

11.2 Diagnosis of the observed failure

Observation	Most likely cause	Design response
Sudden bow near 83 kN with little prior yielding	Stability governs	Increase radius of gyration per unit mass or reduce effective length
Peak between 70 and 83 kN with yielding on the concave face	Eccentricity-driven yield, as predicted	Reduce eccentricity with tighter tolerances or spherical end bearings
Peak below 70 kN	Eccentricity above 2 mm, or strength below typical	Check the measured e and coupon strength; correct the model inputs
Fracture or ovalisation of a tongue below 105 kN	Tension-lug coefficients K and Kn were optimistic	Increase tongue thickness or width; refine K and Kn from the manual's figures
Permanent set in a pin	Pin strength lower than assumed, or bending arm larger than modelled	Use a larger or stronger pin; check the arm
Wall dimple near mid-length	Shell knockdown worse than one third	Increase wall thickness
Crack at the taper shoulder	Stress concentration at the shoulder	Increase shoulder radius

11.3 Redesign levers and success metric

The first-yield load per unit mass at e = 1 mm is 97 kN/kg. The end regions are 67% of the part mass, and the shank mass changes little with eccentricity up to 2 mm (Section 6.2). The principal mass reductions therefore lie in the end regions: a shorter tongue zone and taper, pocketed tongue faces, and a thinner

tongue with a bushing. A second lever is a spherical end bearing to reduce eccentricity, which raises the first-yield load toward the Johnson ceiling. The success criterion is an increase of at least 30% in peak load per unit mass, with all requirements of Section 3.2 still met. One change is made at a time so that each gain can be attributed.

12. Extensions

- **Combined axial and lateral load.** Gimbaling adds a lateral load of about T sin(delta) at the engine, and engine inertia adds more. A combined-load strut and a beam-column analysis are the natural next step.
- **Fatigue and reuse.** Reusable vehicles see many load cycles. The MatWeb extract [9] lists a 6061-T6 fatigue strength of 96.5 MPa at 500 million fully reversed cycles, a starting point for a life estimate with scatter.
- **Low temperature.** MatMatch [18] lists 6061-T6 properties at -196 C, which are higher in strength than at room temperature. A cold test needs a chamber that covers the temperature.
- **Dynamic stiffness.** A modal tap test gives the first axial and lateral frequencies, to compare with the stiffness prediction. NASA SP-8055 [17] describes why structural frequencies must stay clear of propulsion-system frequencies. An early thrust-mount paper [21] discusses the stiffness requirements of thrust mounts.
- **Probabilistic study.** A Monte Carlo analysis that randomises eccentricity, strength and geometry gives a failure-load distribution to compare with the measurement.
- **Full thrust structure.** A four-strut assembly with a thrust ring, including the case of one strut failing, tests load redistribution.
- **Tension case.** The lug equations are formulated for tension (Section 5.6; lug failure modes are described in [22]). A tension test would calibrate the K and Kn coefficients directly.

13. Conclusions

1. The load on a strut is derived from engine thrust, strut count and strut angle: 224.8 kN at full scale and 42.5 kN on the 0.43-scale test article. The ultimate load is 59.5 kN.
2. A two-variable sweep over tube outside diameter and wall thickness gives a shank of 44.0 mm outside diameter and 2.5 mm wall. It is 36% lighter than the lightest solid round bar meeting the same criteria, and buckling governs for eccentricities up to 2 mm.
3. With the end fittings, the part has a mass of 0.805 kg. The end regions are 67% of that mass, so the end regions, not the shank, are the main mass-reduction opportunity.
4. The tongues, holes and pins are sized so that the column fails first. The predicted peak load is 70 to 83 kN, which is 1.65 to 1.96 times the limit load, with yielding on the concave face at mid-length followed by lateral bowing.
5. The predicted axial stiffness is 61.9 kN/mm for the bar model. The measured value will be lower because pin, hole and fork compliance are not in the model.
6. The test plan, FEA specification, correlation table and diagnosis method are defined before testing, so the comparison of hand calculation, FEA and measurement answers the research question directly.

Limitations of the present analysis are stated where they arise: the load case is axial thrust only; the strut count, angle and scale are representative values and not vehicle data; the lug coefficients K and Kn

are conservative placeholders to be replaced from the manual's figures; the pin strengths are manual example values to be replaced by certificate values; and the tension lug equations are applied to a compression test as a conservative bound.

14. Attachments and reproduction

All numbers in this report are produced by the four scripts in Appendices A to D. Run them in this order in one directory: model.py (imported by the others), sweep_tube.py (sizing tables), design_and_prediction.py (design, margins, mass, stiffness and predictions; writes design.json), and figures.py (reads design.json and writes the four figures).

File	Content
model.py	Material data, load definition, tube section, column, secant, shell-buckling, lug and pin functions
sweep_tube.py	Two-variable sizing sweep and sensitivity tables
design_and_prediction.py	Selected shank, end-fitting search, margins, mass, stiffness and test predictions
figures.py	Figures 1 to 4
fig1_free_body_diagrams.png	Load path, eccentric strut and pin-joint free-body diagrams
fig2_geometry.png	Strut geometry
fig3_sizing_map.png	Feasible region and active criteria
fig4_prediction.png	Predicted load against lateral offset

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Appendix A. model.py

```
"""
Common model: 6061-T6 tubular strut with integral clevis-tongue ends, loaded in axial
compression through pins.
Units: N, mm, MPa, kg.  Imported by the sweep, end-fitting and prediction scripts.
"""
```

```
import numpy as np

# ---------- material 6061-T6 ----------
E, G, rho = 68.9e3, 26.0e3, 2.70e-6            # MPa, MPa, kg/mm^3
Fty_d, Ftu_d = 240.0, 290.0                    # design (minimum) yield / ultimate,
MPa
Fty_t, Ftu_t = 276.0, 310.0                    # typical values (used for test
prediction only)
Fsu, Fbru, Fbry = 207.0, 607.0, 386.0          # shear; bearing ultimate / yield (e/D
= 2.0)
# ---------- pin steel (4130 values from Air Force Stress Manual worked example)
----------
ksi = 6.89476
Ftu_P, Fty_P, Fsu_P = 125*ksi, 103*ksi, 82*ksi
kbP = 1.56                                      # plastic bending coefficient,
ductile pin
# ---------- factors and criteria ----------
FS_y, FS_u, MS_buck_min = 1.25, 1.40, 0.10
K = 1.0                                         # pinned-pinned effective length
factor
L = 400.0                                       # pin-centre to pin-centre length, mm
e_ecc = 1.0                                     # load eccentricity used in analysis,
mm
# ---------- load definition ----------
T_full, D_eng = 845e3, 920.0                    # Merlin 1D sea-level thrust (N) and
diameter (mm)
N_str, alpha = 4, np.radians(20.0)
lam = L/D_eng
P_full = T_full/(N_str*np.cos(alpha))
P_L = lam**2*P_full                             # limit load on the test strut
P_y, P_u = FS_y*P_L, FS_u*P_L

# ---------- round tube section ----------
def tube(D, t):
    d = D - 2*t
    A = np.pi/4*(D**2 - d**2); I = np.pi/64*(D**4 - d**4); r = np.sqrt(I/A)
    return A, I, r, D/2

# ---------- column models ----------
def Cc(Fty): return np.pi*np.sqrt(2*E/Fty)
def P_crit(A, r, Fty=Fty_d):
    S = K*L/r
    sig = np.where(S >= Cc(Fty), np.pi**2*E/S**2, Fty - Fty**2/(4*np.pi**2*E)*S**2)
    return sig*A, S
def phi(P, A, r): return K*L/(2*r)*np.sqrt(P/(A*E))
def sigma_secant(P, A, r, c, e=e_ecc):
    f = phi(P, A, r)
    return np.where(f < np.pi/2, P/A*(1 + e*c/r**2/np.cos(np.minimum(f, 1.55))),
np.inf)
def delta_secant(P, A, r, e=e_ecc):
    f = phi(P, A, r)
    return np.where(f < np.pi/2, e*(1/np.cos(np.minimum(f, 1.55)) - 1), np.inf)
def sigma_local(D, t):                          # classical axial shell buckling
stress, 0.605 E t / R (R = mean radius)
    return 0.605*E*t/((D - t)/2)
```

```
def first_yield(A, r, c, e, Fty=Fty_t):
    Pc, _ = P_crit(A, r, Fty); Ps = np.linspace(1, 0.9999*Pc, 60000)
    sg = sigma_secant(Ps, A, r, c, e)
    return Ps[np.argmax(sg >= Fty)] if np.any(sg >= Fty) else Ps[-1]     # no yield
before the Johnson load -> ceiling

# ---------- lug and pin (Air Force Stress Manual, double shear, axial) ----------
def pin_capacity(Dp, t_fork, t_tongue, g):
    arm = t_fork/2 + t_tongue/4 + g                         # eq. 9-13
    return dict(arm=arm,
                ult_bend = 0.1963*kbP*Dp**3*Ftu_P/arm,       # eq. 9-15
                elastic  = 0.1963*Dp**3*Fty_P/arm,
                shear    = 1.571*Dp**2*Fsu_P)                # eq. 9-12
def tongue_capacity(Dp, e, w, t, Kn=0.74, K_b=1.33, Ftu=Ftu_d, Fty=Fty_d):
    Fb = min(K_b*Ftu, 1.304*K_b*Fty, Fbru, 1.304*Fbry) if e/Dp >= 1.5 else np.nan
    return dict(bearing=Fb*Dp*t, net=Kn*Ftu*(w - Dp)*t)
```

Appendix B. sweep_tube.py

```
"""Two-variable sizing sweep (outside diameter D, wall t) for the round tubular strut
shank."""
import numpy as np, json
from model import *

Ds = np.arange(28.0, 60.01, 0.5); ts = np.arange(1.5, 12.01, 0.25)
Dg, tg = np.meshgrid(Ds, ts, indexing="ij")

def evaluate(D, t, e=e_ecc, D_over_t_max=25.0, t_min=2.0):
    A, I, r, c = tube(D, t)
    u_y = sigma_secant(P_y, A, r, c, e)/Fty_d                  # yield at 1.25 x
limit
    u_u = sigma_secant(P_u, A, r, c, e)/Ftu_d                  # ultimate at 1.4 x
limit
    Pc, S = P_crit(A, r)
    u_b = (1 + MS_buck_min)*P_u/Pc                             # buckling margin >=
10 %
    u_l = 3*Fty_d/sigma_local(D, t)                            # classical local
buckling stress >= 3 x yield
    ok = (u_y <= 1) & (u_u <= 1) & (u_b <= 1) & (u_l <= 1) & (D/t <= D_over_t_max) &
(t >= t_min)
    return ok, A, dict(yield_=u_y, ult=u_u, buck=u_b, local=u_l)

def optimum(e=e_ecc, D_max=60.0, **kw):
    ok, A, u = evaluate(Dg, tg, e, **kw); ok &= (Dg <= D_max)
    m = np.where(ok, A, np.inf); i = np.unravel_index(np.argmin(m), m.shape)
    ut = {k: float(v[i]) for k, v in u.items()}
    return Dg[i], tg[i], A[i], rho*A[i]*L, ut, max(ut, key=ut.get)

if __name__ == "__main__":
    print(f"lambda={lam:.4f}  P_full={P_full/1e3:.1f} kN  P_L={P_L/1e3:.2f}
P_y={P_y/1e3:.2f}  P_u={P_u/1e3:.2f} kN")
    print("\n e(mm) Dmax(mm) | OD(mm) wall(mm)  A(mm2) mass(kg) | util yield ult buck
local | governing   [D/t<=25, t>=2.0]")
    rows = {}
    for e in (0.5, 1.0, 2.0, 5.0):
```

```
        for Dm in (40, 50, 60):
            D, t, A, m, ut, gov = optimum(e, Dm)
            print(f" {e:4.1f}   {Dm:4d}    | {D:5.1f}  {t:5.2f}   {A:6.1f}  {m:6.3f}
| {ut['yield_']:.2f} {ut['ult']:.2f} {ut['buck']:.2f} {ut['local']:.2f} | {gov}")
            rows[f"e{e}_D{Dm}"] = dict(D=float(D), t=float(t), A=float(A),
mass=float(m), gov=gov)
    json.dump(rows, open("sweep_tube_results.json", "w"), indent=1)
    print("\nSensitivity to the wall-slenderness bound (e = 1 mm, OD <= 50 mm):")
    for dt in (15, 20, 25, 30):
        D, t, A, m, ut, gov = optimum(1.0, 50, D_over_t_max=dt)
        print(f" D/t<={dt}: OD={D:.1f} t={t:.2f} A={A:.0f} mm2 mass={m:.3f} kg
util(y,u,b)=({ut['yield_']:.2f},{ut['ult']:.2f},{ut['buck']:.2f})  governing: {gov}")
    for Dsol in np.arange(15, 40, 0.5):                      # lightest solid round
bar, for comparison
        A = np.pi/4*Dsol**2; r = Dsol/4; c = Dsol/2; Pc, _ = P_crit(A, r)
        if sigma_secant(P_y, A, r, c)/Fty_d <= 1 and sigma_secant(P_u, A, r, c)/Ftu_d
<= 1 and (1 + MS_buck_min)*P_u/Pc <= 1:
            print(f"lightest solid round bar: D={Dsol} mm, A={A:.0f} mm2,
mass={rho*A*L:.3f} kg"); break
```

Appendix C. design_and_prediction.py

```
"""Baseline selection, end-fitting sizing, mass/stiffness, and test predictions for
the tubular strut."""
import numpy as np, json
from model import *
from sweep_tube import optimum, evaluate, Dg, tg      # sweep helpers

print("\n=== SENSITIVITY TO WALL-SLENDERNESS BOUND (e=1 mm, OD<=50) ===")
for dt in (15, 20, 25, 30):
    D, t, A, m, ut, gov = optimum(1.0, 50, D_over_t_max=dt)
    print(f" D/t<={dt}: OD={D:.1f} t={t:.2f} A={A:.0f} mass={m:.3f} kg
util(y,u,b)=({ut['yield_']:.2f},{ut['ult']:.2f},{ut['buck']:.2f}) gov={gov}")

# ---------------- baseline shank (D/t <= 20, OD <= 50) ----------------
D, t, A_opt, m_opt, ut, gov = optimum(1.0, 50, D_over_t_max=20)
D, t = 44.0, 2.5                                      # sweep optimum OD 44.0; wall
2.25 rounded up to 2.5 mm for boring tolerance
A, I, r, c = tube(D, t); d = D - 2*t
print(f"\nBASELINE SHANK: OD={D} wall={t} ID={d}  A={A:.1f} mm2  I={I:.0f} mm4
r={r:.3f} mm  S={K*L/r:.2f}  Cc(design)={Cc(Fty_d):.2f}")
Pc_d, S = P_crit(A, r); Pc_t, _ = P_crit(A, r, Fty_t)
print(f" Johnson Pcr design={Pc_d/1e3:.1f} kN (MS vs 1.1Pu {Pc_d/(1.1*P_u)-1:+.2f});
typical ceiling={Pc_t/1e3:.1f} kN; Euler ideal={np.pi**2*E*I/(K*L)**2/1e3:.1f} kN")
for nm, P, F in (("limit", P_L, None), ("1.25 x limit", P_y, Fty_d), ("1.4 x limit",
P_u, Ftu_d)):
    f = float(phi(P, A, r)); sg = float(sigma_secant(P, A, r, c)); dl =
float(delta_secant(P, A, r))
    print(f" {nm:13s} P={P/1e3:6.2f} kN  P/A={P/A:6.1f}  phi={f:.4f}
sec={1/np.cos(f):.4f} delta={dl:.3f} mm  sigma_max={sg:.1f} MPa" + (f"  MS={F/sg-
1:+.2f}" if F else ""))
print(f" local buckling: classical sigma_cl={sigma_local(D,t):.0f} MPa =
{sigma_local(D,t)/Fty_d:.1f} x design yield; D/t={D/t:.0f}")
print(f" shank mass per mm of tube = {rho*A:.5f} kg/mm")
```

```
# ---------------- end fittings ----------------
P_end, P_pin_el = 1.25*Pc_t, 1.10*Pc_t
print(f"\nEND REQUIREMENTS: strut ceiling {Pc_t/1e3:.1f} kN -> ends >=
{P_end/1e3:.1f} kN, pin elastic >= {P_pin_el/1e3:.1f} kN")
t_fork, g = 10.0, 0.5
best = None
for Dh in (50.0, 56.0, 60.0, 63.5):                    # head bar stock diameter
(tongue is milled from it)
    for Dp in (16, 18, 20, 22, 25, 28, 30):
        for tl in np.arange(10, 0.8*Dh+0.1, 1.0):
            w = np.sqrt(Dh**2 - tl**2); e = float(np.ceil(1.6*Dp/5)*5)
            pn = pin_capacity(Dp, t_fork, tl, g); tc = tongue_capacity(Dp, e, w, tl)
            ok = (pn['elastic'] >= P_pin_el) and (pn['ult_bend'] >= P_end) and
(pn['shear'] >= P_end) and (tc['bearing'] >= P_end) and (tc['net'] >= P_end)
            if ok:
                vol = tl*w*(e+20)
                if best is None or vol < best[0]: best = (vol, Dh, Dp, tl, w, e, pn,
tc)
vol, Dh, Dp, tl, w, e_h, pn, tc = best
print(f" minimum-volume tongue: head stock D={Dh}, pin D={Dp} mm, tongue
thickness={tl:.0f}, chord width={w:.1f}, e={e_h:.0f} (e/D={e_h/Dp:.2f},
a={e_h-Dp/2:.1f})")
Dp, tl, e_h = float(Dp), float(tl), float(e_h); w = np.sqrt(Dh**2 - tl**2)
pn = pin_capacity(Dp, t_fork, tl, g); tcd = tongue_capacity(Dp, e_h, w, tl); tct =
tongue_capacity(Dp, e_h, w, tl, Ftu=Ftu_t, Fty=Fty_t)
print(f" pin: arm={pn['arm']:.2f} mm  ult bending={pn['ult_bend']/1e3:.0f}
elastic={pn['elastic']/1e3:.0f}  shear={pn['shear']/1e3:.0f} kN")
print(f" tongue (design min): bearing={tcd['bearing']/1e3:.1f}
net={tcd['net']/1e3:.1f} kN ; typical: bearing={tct['bearing']/1e3:.1f}
net={tct['net']/1e3:.1f} kN")
for nm, cap, req in (("tongue bearing", tcd['bearing'], P_end), ("tongue net
section", tcd['net'], P_end), ("pin shear", pn['shear'], P_end), ("pin ult bending",
pn['ult_bend'], P_end), ("pin elastic", pn['elastic'], P_pin_el)):
    print(f"   {nm:19s} MS vs requirement = {cap/req-1:+.2f}   | vs ultimate design
load {cap/P_u-1:+.2f}")
print(f" tear-out sanity (2 planes, Fsu={Fsu}):
{2*Fsu*tl*(e_h-Dp/2*np.cos(np.radians(40)))/1e3:.0f} kN")

# ---------------- geometry, mass, stiffness ----------------
L_tot = L + 2*e_h; L_tongue = e_h + 20.0; L_taper = 30.0; L_tube = L_tot -
2*(L_tongue + L_taper)
R = Dh/2; zt = tl/2
A_tongue = 2*(zt*np.sqrt(R**2 - zt**2) + R**2*np.arcsin(zt/R)); A_head =
np.pi*Dh**2/4; A_tube_end = np.pi*D**2/4
vol_tube = A*L_tube
vol_taper = 2*(np.pi*L_taper/12)*(Dh**2 + Dh*D + D**2)             # frustum, solid
vol_tongue = 2*A_tongue*L_tongue - 2*np.pi/4*Dp**2*tl
mass = rho*(vol_tube + vol_taper + vol_tongue)
print(f"\nGEOMETRY: total length {L_tot:.0f}, tongue zone {L_tongue:.0f}, taper
{L_taper:.0f}, tube {L_tube:.0f} mm; head stock {Dh} mm; tube OD {D} wall {t}")
print(f"MASS = {mass:.3f} kg  (tube {rho*vol_tube:.3f}, tapers {rho*vol_taper:.3f},
tongues {rho*vol_tongue:.3f})")
A_net_h = tl*(w - Dp)
comp = L_tube/(E*A) + 2*L_taper/(E*(A_head+A_tube_end)/2) + 2*20/(E*A_net_h) +
2*(L_tongue-20)/(E*A_tongue)
```

```
print(f"AXIAL STIFFNESS: EA/L(tube, L=400)={E*A/L/1e3:.1f} kN/mm ; segmented-bar
model={1/comp/1e3:.1f} kN/mm")

# ---------------- predictions (typical properties) ----------------
print("\nPREDICTION (typical): ceiling (Johnson) = %.1f kN" % (Pc_t/1e3))
pred = {}
for ee in (0.5, 1.0, 2.0):
    fy = first_yield(A, r, c, ee); pred[ee] = fy
    print(f"  e={ee}: first yield = {fy/1e3:.1f} kN  ({fy/P_L:.2f} x limit)   delta
at limit = {float(delta_secant(P_L,A,r,ee)):.3f} mm")
print(f"  planning load for machine = 1.5 x ceiling = {1.5*Pc_t/1e3:.0f} kN =
{1.5*Pc_t/4448.2:.0f} kip ; fork bearing at that load = {1.5*Pc_t/(Dp*2*t_fork):.0f}
MPa")
print(f"  peak/mass at e=1: {pred[1.0]/1e3/mass:.0f} kN/kg")
json.dump(dict(D=D,t=t,Dh=Dh,Dp=Dp,tl=tl,e_h=e_h,mass=float(mass),ceiling=float(Pc_t)
,first_yield={str(k):float(v) for k,v in pred.items()}), open("design.json","w"),
indent=1)
```

Appendix D. figures.py

```
"""Figures: load-path / free-body diagrams, geometry drawing, sizing map, predicted
load-deflection curves."""
import numpy as np, json
import matplotlib; matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.patches import Rectangle, FancyArrowPatch, Circle, Polygon
from model import *
import sweep_tube as sw
dsg = json.load(open("design.json")); D, t, Dh, Dp, tl, e_h = dsg["D"], dsg["t"],
dsg["Dh"], dsg["Dp"], dsg["tl"], dsg["e_h"]
plt.rcParams.update({"font.size": 9})
def arr(ax, p, q, c="red", lw=2, **k): ax.add_patch(FancyArrowPatch(p, q,
arrowstyle="-|>", mutation_scale=14, color=c, lw=lw, **k))

# ---- Fig 1: free-body diagrams ----
fig, ax = plt.subplots(1, 3, figsize=(13, 4.6))
a = ax[0]; a.set_xlim(0, 10); a.set_ylim(0, 8); a.axis("off"); a.set_title("(a)
Vehicle-level load path")
a.add_patch(Rectangle((1, 6.2), 8, 0.6, fc="#dddddd", ec="k")); a.text(5, 6.9,
"thrust frame / tank bulkhead", ha="center")
a.add_patch(Polygon([[4.3, 2.0], [5.7, 2.0], [6.1, 0.6], [3.9, 0.6]], fc="#bbbbbb",
ec="k")); a.text(5, 0.2, "engine", ha="center")
for x0 in (1.8, 3.4, 6.6, 8.2):
    a.plot([x0, 4.6 if x0 < 5 else 5.4], [6.2, 2.0], "k-", lw=3)
arr(a, (5, 1.2), (5, 5.2), "tab:blue", 3); a.text(5.2, 4.0, "T", color="tab:blue",
fontsize=12)
a.text(0.2, 1.2, "N = 4 struts\nat angle alpha = 20 deg\nP = T / (N cos alpha)",
fontsize=9)
b = ax[1]; b.set_xlim(0, 10); b.set_ylim(0, 8); b.axis("off"); b.set_title("(b)
Strut: two-force member, eccentric load")
b.plot([5, 5], [1, 7], "k-", lw=7)
xs = np.linspace(1, 7, 50); b.plot(5 + 0.5*np.sin(np.pi*(xs-1)/6), xs, "b:", lw=2)
arr(b, (5.8, 7.6), (5.8, 7.05), "red"); arr(b, (5.8, 0.4), (5.8, 0.95), "red")
b.text(6.2, 7.5, "P", color="red", fontsize=12); b.text(6.2, 0.2, "P", color="red",
fontsize=12)
```

```
b.annotate("", xy=(5, 7.3), xytext=(5.8, 7.3), arrowprops=dict(arrowstyle="<->"));
b.text(5.3, 7.75, "e", fontsize=11)
b.annotate("", xy=(5, 4), xytext=(5.5, 4), arrowprops=dict(arrowstyle="<->",
color="b")); b.text(5.6, 4, "delta", color="b")
b.annotate("", xy=(3.2, 1), xytext=(3.2, 7), arrowprops=dict(arrowstyle="<->"));
b.text(0.2, 4, "L = 400 mm\n(pin to pin)", fontsize=9)
b.text(8.0, 5.5, "sigma_max = P/A [1 + (e c/r^2)\n sec(KL/2r sqrt(P/AE))]",
fontsize=8, ha="center")
c = ax[2]; c.set_xlim(0, 12); c.set_ylim(0, 8); c.axis("off"); c.set_title("(c) Pin
joint: double shear")
c.add_patch(Rectangle((1, 3.2), 4.5, 1.6, fc="#cfe2f3", ec="k")); c.text(2.8, 4.0,
"strut tongue\n(thickness t_l)", ha="center", va="center")
for y0 in (4.9, 2.3): c.add_patch(Rectangle((4, y0), 5, 0.9, fc="#eeeeee", ec="k"))
c.text(7.8, 5.35, "fork plate (t1)", ha="center"); c.text(7.8, 2.75, "fork plate
(t1)", ha="center")
c.add_patch(Rectangle((4.6, 2.3), 0.6, 3.5, fc="#999999", ec="k")); c.text(4.9, 1.8,
"pin", ha="center")
arr(c, (0.2, 4.0), (1.0, 4.0), "red"); c.text(0.1, 4.4, "P", color="red",
fontsize=12)
arr(c, (9.8, 5.35), (9.1, 5.35), "red"); arr(c, (9.8, 2.75), (9.1, 2.75), "red");
c.text(9.9, 5.6, "P/2", color="red"); c.text(9.9, 3.0, "P/2", color="red")
c.text(6.0, 0.7, "pin bending arm = t1/2 + t_l/4 + g", ha="center")
plt.tight_layout(); plt.savefig("fig1_free_body_diagrams.png", dpi=170); plt.close()

# ---- Fig 2: geometry ----
L_tot = L + 2*e_h; Lt = e_h + 20.0; Lk = 30.0
fig, ax = plt.subplots(figsize=(13, 3.6)); ax.set_aspect("equal"); ax.axis("off")
xs_ = [0, Lt, Lt+Lk, L_tot-Lt-Lk, L_tot-Lt, L_tot]
ax.add_patch(Polygon([[xs_[0], -tl/2], [xs_[1], -tl/2], [xs_[1], -Dh/2], [xs_[2], -
D/2], [xs_[3], -D/2], [xs_[4], -Dh/2], [xs_[4], -tl/2], [xs_[5], -tl/2], [xs_[5],
tl/2], [xs_[4], tl/2], [xs_[4], Dh/2], [xs_[3], D/2], [xs_[2], D/2], [xs_[1], Dh/2],
[xs_[1], tl/2], [xs_[0], tl/2]], fc="#cfe2f3", ec="k"))
ax.add_patch(Rectangle((xs_[2]+0.0, -(D/2-t)), xs_[3]-xs_[2], D-2*t, fc="white",
ec="k", ls="--"))
for xc in (e_h, L_tot - e_h):
    ax.add_patch(Circle((xc, 0), 2.0, fc="k"))
    ax.plot([xc, xc], [-Dh/2-4, Dh/2+4], "k-.", lw=0.6)
ax.annotate("", xy=(e_h, -Dh/2-8), xytext=(L_tot-e_h, -Dh/2-8),
arrowprops=dict(arrowstyle="<->")); ax.text(L_tot/2, -Dh/2-14, f"pin centre distance
L = {L:.0f} mm", ha="center")
ax.annotate("", xy=(0, Dh/2+8), xytext=(L_tot, Dh/2+8),
arrowprops=dict(arrowstyle="<->")); ax.text(L_tot/2, Dh/2+12, f"overall length
{L_tot:.0f} mm", ha="center")
ax.text(L_tot/2, 0, f"tube OD {D:.0f} x wall {t:.1f} (bored)", ha="center",
va="center", fontsize=9)
ax.text(Lt/2+5, Dh/2+0, f"tongue {tl:.0f} mm thick\npin hole {Dp:.0f} H7, e =
{e_h:.0f} mm", ha="center", va="top", fontsize=8)
ax.set_xlim(-20, L_tot+20); ax.set_ylim(-Dh/2-22, Dh/2+22); ax.set_title("Strut
geometry, edge view of the tongues (schematic; shoulder radii not drawn)")
plt.tight_layout(); plt.savefig("fig2_geometry.png", dpi=170); plt.close()

# ---- Fig 3: sizing map and constraint utilisation ----
ok, A_, u = sw.evaluate(sw.Dg, sw.tg, 1.0, D_over_t_max=20)
fig, ax = plt.subplots(1, 2, figsize=(12, 4.4))
ax[0].contourf(sw.Dg, sw.tg, ok, levels=[-.5, .5, 1.5], colors=["#f4cccc",
```

```
"#d9ead3"])
cs = ax[0].contour(sw.Dg, sw.tg, rho*A_*L, levels=[0.25, 0.3, 0.35, 0.4, 0.5, 0.6],
colors="gray", linewidths=.7); ax[0].clabel(cs, fmt="%.2f kg")
ax[0].contour(sw.Dg, sw.tg, u["buck"], levels=[1], colors="b", linestyles="--");
ax[0].contour(sw.Dg, sw.tg, np.maximum(u["yield_"], u["ult"]), levels=[1],
colors="r")
ax[0].plot(D, t, "ko"); ax[0].plot(44, 2.25, "k^")
ax[0].set_xlabel("tube outside diameter (mm)"); ax[0].set_ylabel("wall thickness
(mm)"); ax[0].set_ylim(1.5, 8); ax[0].set_xlim(28, 60)
ax[0].set_title("Feasible region (green, e = 1 mm, D/t <= 20)\nblue dashed: buckling
limit, red: strength limit")
Dv = np.arange(30, 60.1, 0.5); tv = np.full_like(Dv, 2.5)
okv, Av, uv = sw.evaluate(Dv, tv, 1.0, D_over_t_max=100)
ax[1].plot(Dv, uv["yield_"], label="yield (1.25 x limit)"); ax[1].plot(Dv, uv["ult"],
label="ultimate (1.4 x limit)"); ax[1].plot(Dv, uv["buck"], label="buckling (MS >=
10%)")
ax[1].axhline(1, color="k", lw=.8); ax[1].set_ylim(0, 2); ax[1].set_xlabel("tube OD
at 2.5 mm wall (mm)"); ax[1].set_ylabel("demand / allowable"); ax[1].legend();
ax[1].set_title("Active criterion along the 2.5 mm wall line")
plt.tight_layout(); plt.savefig("fig3_sizing_map.png", dpi=170); plt.close()

# ---- Fig 4: predicted load vs lateral offset ----
A, I, r, c = tube(D, t); fig, ax = plt.subplots(figsize=(6.4, 4.4))
Pc_t, _ = P_crit(A, r, Fty_t)
for e in (0.5, 1.0, 2.0):
    Ps = np.linspace(0, 0.995*Pc_t, 800); ax.plot(delta_secant(Ps, A, r, e) + e,
Ps/1e3, label=f"e = {e} mm")
ax.axhline(P_L/1e3, color="g", ls=":", label="limit load"); ax.axhline(P_u/1e3,
color="orange", ls="--", label="1.4 x limit")
ax.axhline(Pc_t/1e3, color="k", lw=.7, label="Johnson ceiling (typical)")
ax.set_xlim(0, 8); ax.set_xlabel("mid-length lateral offset (mm)");
ax.set_ylabel("axial load (kN)"); ax.set_title("Predicted load vs lateral offset
(secant model)"); ax.legend(fontsize=8)
plt.tight_layout(); plt.savefig("fig4_prediction.png", dpi=170); plt.close()
print("figures written")
```